

# Kidney Transplantation Surgery

*Adam D. Barlow, Michael L. Nicholson*

## SOURCES OF KIDNEYS FOR TRANSPLANTATION

The usual and most frequent source of kidneys for transplantation has been donation after brain death (DBD), formerly known as donation from a heart-beating cadaveric donor. The increasing worldwide discrepancy between the availability of and the need for kidney allografts<sup>1</sup> has led to the increasing use of alternative sources of organs, including donation after cardiac death (DCD) donors (previously known as non-heart-beating donors) and living donors. The evaluation and selection of donors are discussed in [Chapter 107](#). This chapter discusses surgical aspects of retrieval and transplantation of kidneys.

## DONATION AFTER BRAIN DEATH DONORS

The potential DBD donor is maintained by artificial ventilation in a critical care setting until death has been diagnosed by brainstem death criteria,<sup>2</sup> the consent of the next of kin for donation has been given, and the necessary legal and institutional approvals have been obtained.

In the operating room some dissection is undertaken initially to define any aberrant anatomy and increase the speed and safety of organ retrieval after perfusion (warm dissection). Cannulation of the aorta and inferior vena cava is performed while the heart is still beating. This allows perfusion of the organs with cold preservative solution immediately before cardiac arrest, minimizing warm ischemia. The priorities of the organ retrieval team are influenced by the range of organs being donated. Heart, lung, liver, and pancreas retrieval take priority over kidney retrieval, which may significantly lengthen the ischemic time. The kidneys are removed with a cuff of aorta (Carrel patch) attached to the renal artery, with the maximum achievable length of renal vein, and 10 to 15 cm of ureter. The length of the right renal vein should be maximized by including a portion of inferior vena cava in continuity. Care is taken to avoid damage to polar and other accessory arteries, especially the lower pole artery, which may supply the ureter. Stripping of adventitial tissue from the ureter must also be avoided, because this also may compromise its blood supply.

The kidneys are flushed with ice-cold preservation fluid until the effluent is clear and then are stored for transport in crushed ice or on a perfusion machine (see the discussion of kidney preservation).

## DONATION AFTER CARDIAC DEATH DONORS

Before consensus was reached regarding the definition of brainstem death, DCD donors were the main source of transplant organs. These donors were intensive care unit based and had sustained head injuries or cerebrovascular accidents deemed irrecoverable, but organ retrieval could proceed only after cardiorespiratory death. This changed with the introduction of brainstem death legislation, but the use of DCD kidneys has recently increased again in response to the shortage of

suitable organs for transplantation. An international consensus has defined categories of DCD donors<sup>3</sup> to facilitate legal and ethical discussion and to highlight possible differences in organ viability (see [Chapter 107, Box 107.6](#)). DCD kidneys sustain a period of warm ischemia, the period between cardiac arrest and the time that in situ cold perfusion is started. The duration of ischemia correlates with rates of primary nonfunction, delayed graft function, acute rejection, allograft, and patient survival. The main requirement of organ procurement from DCD donors is therefore to achieve rapid in situ perfusion of the kidneys to limit warm ischemia. This requires an emergency response team of surgeons and transplant coordinators, with considerable on-call and logistic commitments.

DCD donors may be either uncontrolled (Maastricht categories I and II) or controlled (Maastricht categories III through V). In controlled donors, cardiac arrest is expected, and it is therefore possible to reduce the warm ischemia time to only a few minutes because the surgical retrieval team will be on standby. Unexpected donor cardiac arrest may result in prolonged warm ischemia times. The duration of reversible warm ischemia time that the human kidney can sustain is unknown, but DCD kidneys with warm ischemia exceeding 60 minutes are considered by many to be of marginal suitability.

## Donation After Cardiac Death Protocol

Centers involved with DCD donation should adhere to the Maastricht protocol,<sup>4</sup> which includes the following principles:

- Approval by the local medical ethics committee
- Diagnosis of death by doctors who are independent of the transplantation team
- Five-minute rule (after declaration of cardiac death, the body is left untouched for a period of at least 5 minutes before intervention)
- Rapid in situ cooling with use of a catheter inserted into the aorta
- Organ retrieval by standard surgical techniques

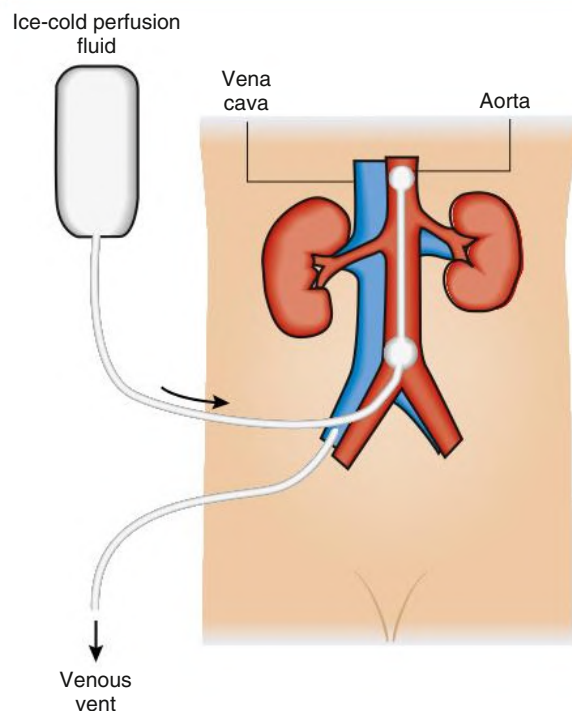
## Uncontrolled Donation After Cardiac Death Donors

After a period of unsuccessful resuscitation, confirmation of cardiac death, and observation of the 5-minute rule, cardiac massage and ventilation with 100% oxygen are recommenced in an attempt to deliver oxygenated blood to the kidneys. A mechanical resuscitation device may be used. In situ renal cooling is achieved by placing a double-balloon, triple-lumen perfusion catheter into the aorta via a femoral artery cut-down ([Fig. 108.1](#)) with instillation of preservation solution. Alternatively, the donor can be moved to an operating room as soon as death has occurred, and the aortic perfusion catheter placed directly at laparotomy rather than via a femoral artery cutdown.

## Controlled Donation After Cardiac Death Donors

For controlled DCD donors, the transplantation team awaits cardiac arrest following withdrawal of life sustaining treatment. This usually

### In Situ Perfusion of Non-Heart-Beating Donor Kidneys



**Fig. 108.1** Technique for in Situ Perfusion of Donation After Cardiac Death Kidneys. A double-lumen, double-balloon arterial catheter is introduced through the femoral artery and the lower balloon inflated at the aortic bifurcation and the upper balloon above the renal arteries. Ice-cold perfusion fluid is introduced and vented through the femoral vein until the effluent becomes clear.

takes place in the theatre anesthetic room or intensive care unit if it is close to the operating theatre. Withdrawal involves cessation of mechanical ventilation, extubation, and cessation of inotropic medications. Following asystole and confirmation of death, the 5-minute rule is observed and the patient is transferred to the operating theatre. A rapid laparotomy is performed and the perfusion catheter is inserted via the aorta or iliac artery.

### LIVING KIDNEY DONORS

In the United States in 2018, 29% of kidney transplants were from living donors<sup>5</sup> compared with 28% in the United Kingdom in 2018–19.<sup>6</sup> After a rapid increase in living donor transplantation at the beginning of this decade, rates have become more static. Superior recipient post-transplantation outcome compared with use of kidneys from deceased donors,<sup>7</sup> the potential for preemptive transplantation before dialysis, and the ability to plan the procedure (allowing optimization of recipient condition) are major advantages and justify continued efforts to expand use of living donors. The medical evaluation of the living donor is discussed in [Chapter 107](#).

#### Preoperative Imaging

Preoperative imaging of living donors confirms the presence of two functioning kidneys, indicates pathology that would preclude donation, and provides anatomic information necessary for planning the procedure. Imaging assumes paramount importance before minimal access donor nephrectomy because of the reduced operative exposure



**Fig. 108.2** Living Donor Preoperative Computed Tomographic Angiogram. (A) Three-dimensional reconstruction of arterial supply. Note the lower pole artery to the right kidney (arrow), which may supply the ureter as well as the lower pole parenchyma. (B) Conventional image showing single artery and vein to the left kidney.

and particular difficulties in the identification of complex vascular anatomy. The location, size, and number of renal arteries and veins needs to be accurately described preoperatively. Angiography combined with excretion urography is now obsolete. For preoperative description of the main renal artery and vein anatomy, magnetic resonance angiography and computed tomographic angiography are comparable,<sup>8</sup> but computed tomographic angiography is more sensitive and specific for complex vascular anatomy and provides excellent correlation between imaging and surgical findings<sup>9</sup> ([Fig. 108.2](#)).

#### Minimal Access (Laparoscopic) Donor Nephrectomy

Living donor nephrectomy was traditionally performed through an open incision, necessitating a prolonged period of recovery. This and the cosmetic implications of a large flank wound may discourage potential donors ([Fig. 108.3](#)). To reduce such disincentives, there was



**Fig. 108.3** Flank wound from open nephrectomy.

### BOX 108.1 Donor Benefits of Minimally Invasive Donor Nephrectomy

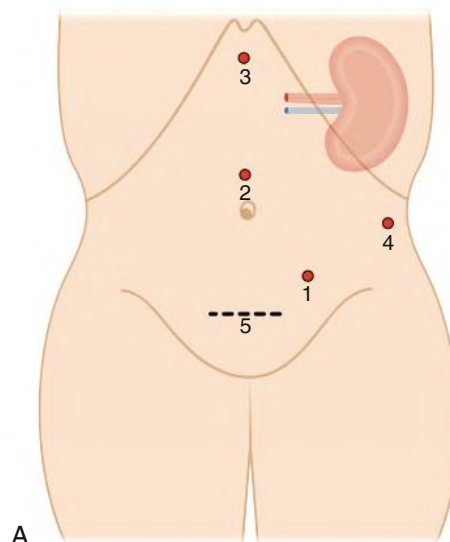
- Shorter incisions
- Less pain
- Shorter hospital stay
- Shorter recovery
- Better cosmetic appearance

a move in the late 1990s and early 2000s toward minimally invasive donor nephrectomy, first performed as a transperitoneal laparoscopic procedure (laparoscopic donor nephrectomy [LapDN]).<sup>10</sup> LapDN is associated with decreased severity and duration of postoperative pain, shorter inpatient stay, quicker return to work and normal activities, and improved cosmetic result compared with open donor nephrectomy<sup>11</sup> (Box 108.1). Furthermore, the overall societal cost of LapDN is lower,<sup>12</sup> and recipient quality-of-life scores are higher.<sup>13</sup> The procedure is, however, technically demanding, and there is potential for damage to the kidney parenchyma, vessels, and ureter during dissection. It takes longer than open nephrectomy and exposes the allograft to a longer period of warm ischemia, albeit only of a few minutes.<sup>11,14</sup>

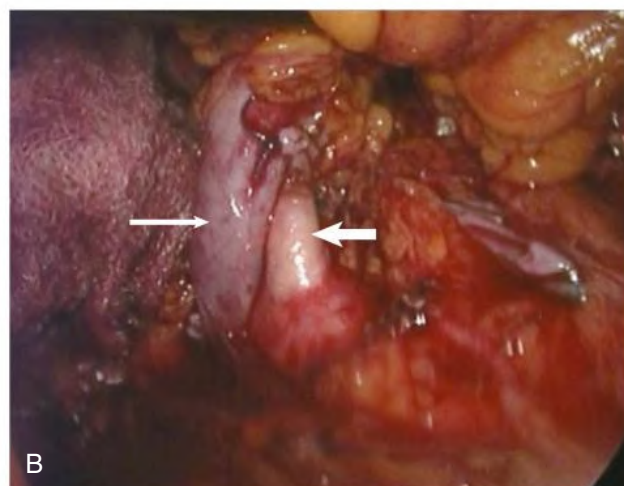
Nevertheless, retrospective data suggest that minimal access donor nephrectomy not only offers postoperative advantages to the donor but also increases the number of transplants performed by reducing donor disincentives; estimates range from a 25% to a 100%<sup>15</sup> increase in transplantation activity. The widespread introduction of LapDN at the beginning of the 2000s saw an initial dramatic increase in the number of live kidney donors. However, rates have been static in both the United States and the United Kingdom over the last 5 years, suggesting that we may have seen the maximum benefits of this effect. Three minimal-access approaches have been described: transperitoneal, extraperitoneal, and hand-assisted living donor nephrectomy.

### Transperitoneal LapDN

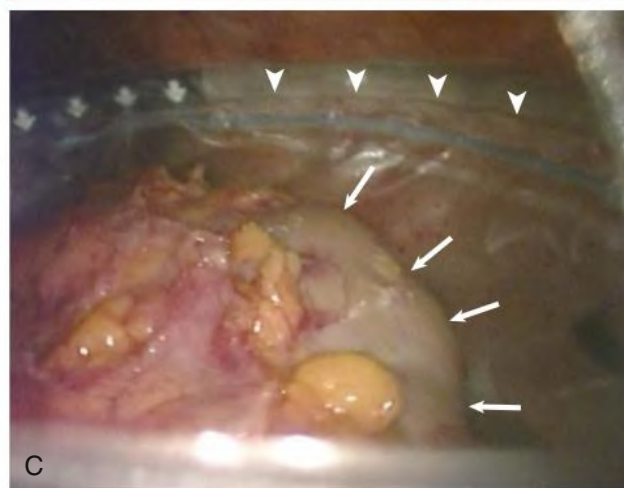
Pneumoperitoneum is established, and four laparoscopic ports are usually required (Fig. 108.4). After laparoscopic dissection a Pfannenstiel or iliac fossa incision is made through which the kidney is brought out within an endoscopy retrieval bag after control and division of the artery vein and ureter.



**A**



**B**



**C**

**Fig. 108.4** Technique for Laparoscopic Donor Nephrectomy. (A) Positions for four laparoscopic ports (1–4) and Pfannenstiel incision (5), through which the kidney is removed. (B) Intraoperative view showing left renal artery (*short arrow*) and vein (*long arrow*) prepared for control and division. (C) Intraoperative view showing the kidney (*arrows* mark lateral margin) placed in the endoscopic retrieval bag (*arrowheads* mark the edge of the bag).



### Hand-Assisted LapDN

The hand-assisted technique allows tactile sense to facilitate dissection, retraction, and exposure. It is said to be easier to learn and can be safely and efficiently performed by surgeons with less laparoscopic experience. The hand-assist device allows the operator's nondominant hand to enter the abdomen through an airtight system.

### Retroperitoneoscopic Operative Technique

The retroperitoneal approach avoids breaching the peritoneum, displays the kidney anatomy in a very different manner, and may be easier for retrieving the full length of the vessels, especially on the right side. The disadvantage is that a more limited operating space is available than with the transperitoneal or hand-assisted laparoscopic techniques.

### Robotic Live Donor Nephrectomy

All of the earlier minimally invasive techniques can be undertaken using robotic assistance, and this has been adopted by a number of centers worldwide. However, there is no high-level evidence of clinical benefit over laparoscopic-assisted approaches, and it is difficult to justify the additional costs and infrastructure required, given the excellent outcomes of the current minimally invasive techniques.

### Contraindications to Minimal Access Donor Nephrectomy

There are no absolute contraindications other than those applying to the open operation. The relative contraindications are dictated by donor factors and the experience of the surgeon. The donor must be fit for anesthesia, including the physiologic stress of pneumoperitoneum. Obesity is a relative contraindication for both open and laparoscopic surgery, and the hand-assisted approach may be better suited in such patients. Previous abdominal surgery is a further relative contraindication because of the potential for adhesions. Multiplicity of renal vessels should not hinder LapDN.

### Effect of Pneumoperitoneum

Transient intraoperative oliguria secondary to decreased renal blood flow is a frequent occurrence during laparoscopic procedures. Proposed mechanisms include decreased cardiac output, renal vein compression, ureteral obstruction, renal parenchymal compression, and systemic hormonal effects. Intracranial pressure increases during pneumoperitoneum, with release of vasoconstrictor agents that decrease renal blood flow. Use of a lower pressure reduces the adverse effects of pneumoperitoneum on kidney perfusion. In donor nephrectomy, impaired renal blood flow may compromise early allograft function and compound the damaging effects of warm and cold ischemia and operative manipulation of the kidney. The pioneers of LapDN used high volumes of crystalloid preoperatively and intraoperatively to maintain kidney perfusion in the presence of pneumoperitoneum. However, this was associated with episodes of unilateral pulmonary edema in the dependent lung, and we now recommend no additional volume loading in the donor beyond standard perioperative fluid replacement. The revised protocol has not apparently reduced the likelihood of good graft function.

### Graft Function and Acute Rejection

There is no consistent evidence that graft function differs among kidneys retrieved by open, laparoscopic, or hand-assisted donor nephrectomy. The exception is that rates of delayed graft function and acute rejection may be higher in pediatric recipients, especially the 0- to 5-year age group.

Pretransplantation ischemia could, in theory, render the donor kidney more immunogenic by inducing major histocompatibility complex (MHC) class II expression. However, despite the longer warm

ischemia time, acute rejection rates and severity of rejection are not higher in laparoscopic than in openly retrieved living donor kidneys.

### Technical Issues

Ureteral ischemia was more common in early experience of LapDN but can be avoided if care is taken to ensure that sufficient periureteral tissue is taken and that the dissection does not occur too close to the renal pelvis.

Multiple arteries need not be a barrier to successful use of grafts from laparoscopic donors. In open donor nephrectomy, the right kidney is retrieved in 20% to 30% of procedures, whereas LapDN uses the right kidney in fewer than 10%,<sup>16</sup> reflecting concern over the operative safety of the right-sided laparoscopic operation, principally the difficulties involved in obtaining an adequate length of renal vein. It has been argued that this practice has led to compromise of the principle that the better kidney should remain with the donor.

### Postoperative Recovery

After uneventful open nephrectomy, the donor can expect to be discharged from the hospital in 5 or 6 days and is able to return to work after 3 to 6 weeks, although return to work has been shown to be 2 to 3 weeks later in open nephrectomy compared with LapDN.<sup>11</sup> After LapDN the donor usually leaves the hospital in 1 to 2 days and can return to work in 3 to 6 weeks.

### Choice of Donor Operative Technique

The choice of operative procedure depends on the local expertise of the surgeons. There is accumulating evidence that the laparoscopic operation removes some of the disincentives to donation, and this approach has now been widely adopted around the world.

## KIDNEY PRESERVATION

Preservation of deceased donor organs is crucial to allow time for matching, sharing of organs, and preparation of the recipient. Damage from hypothermia and reperfusion must be minimized. There is little standardization of the type of preservation solution used. Marshall's hyperosmolar citrate solution and histidine-tryptophan-ketoglutarate are popular choices in Europe, but University of Wisconsin solution is more commonly used in the United States, where extended preservation times are more often required.

Organs can be preserved by cold storage (kept in crushed ice after flushing with preservation solution) or by machine-driven pulsatile perfusion. The proposed benefits of machine perfusion come from allowing aerobic function through provision of oxygen and substrate and removal of metabolic end-products. Although hypothermic machine perfusion has been used for many years, there is still no consensus about its superiority to cold storage nor about the best perfusion parameters. However, recent meta-analyses have suggested benefits of machine perfusion in improved delayed graft function rates and graft survival.<sup>17,18</sup> This effect seems to be similar for kidneys from both DCD and DBD donors and to have cost benefits over static cold storage.<sup>18</sup>

Further developments in hypothermic machine perfusion have led to the addition of oxygen to the perfusion fluid. Clinical trials are ongoing to assess the benefits of hypothermic oxygenated perfusion over both static cold storage and standard hypothermic machine perfusion, with one recent RCT suggesting a reduction in severe complications and graft failure with the addition of oxygen to hypothermic machine perfusion.<sup>19</sup>

A novel approach to kidney preservation, only recently introduced into clinical practice, is *ex vivo* normothermic perfusion. Early results

are promising,<sup>20</sup> but the technique needs to be studied in larger clinical trials, one of which is ongoing in the United Kingdom.<sup>21</sup>

Decisions on the use of a kidney from a marginal donor can be supported by data from machine or normothermic perfusion; high perfusion pressures are associated with primary nonfunction and delayed graft function.

## KIDNEY TRANSPLANTATION PROCEDURE

The transplanted kidney is placed heterotopically in one or another iliac fossa. The right side is usually preferred for first-time kidney transplant as the external iliac vessels, particularly the vein, lie more superficially than on the left. The inferior epigastric vessels are ligated, as is the round ligament of the uterus in female patients. Occasionally the inferior epigastric artery may be preserved and used for revascularization of small polar arteries. In male patients the spermatic cord is mobilized and preserved. The peritoneum should not be breached but instead swept superiorly to reveal the extraperitoneal bed into which the transplanted kidney will be placed. The external iliac blood vessels are then mobilized, with care taken to meticulously ligate all the associated lymphatic channels to reduce the risk for posttransplantation lymphatic leak.

### Vascular Anastomosis

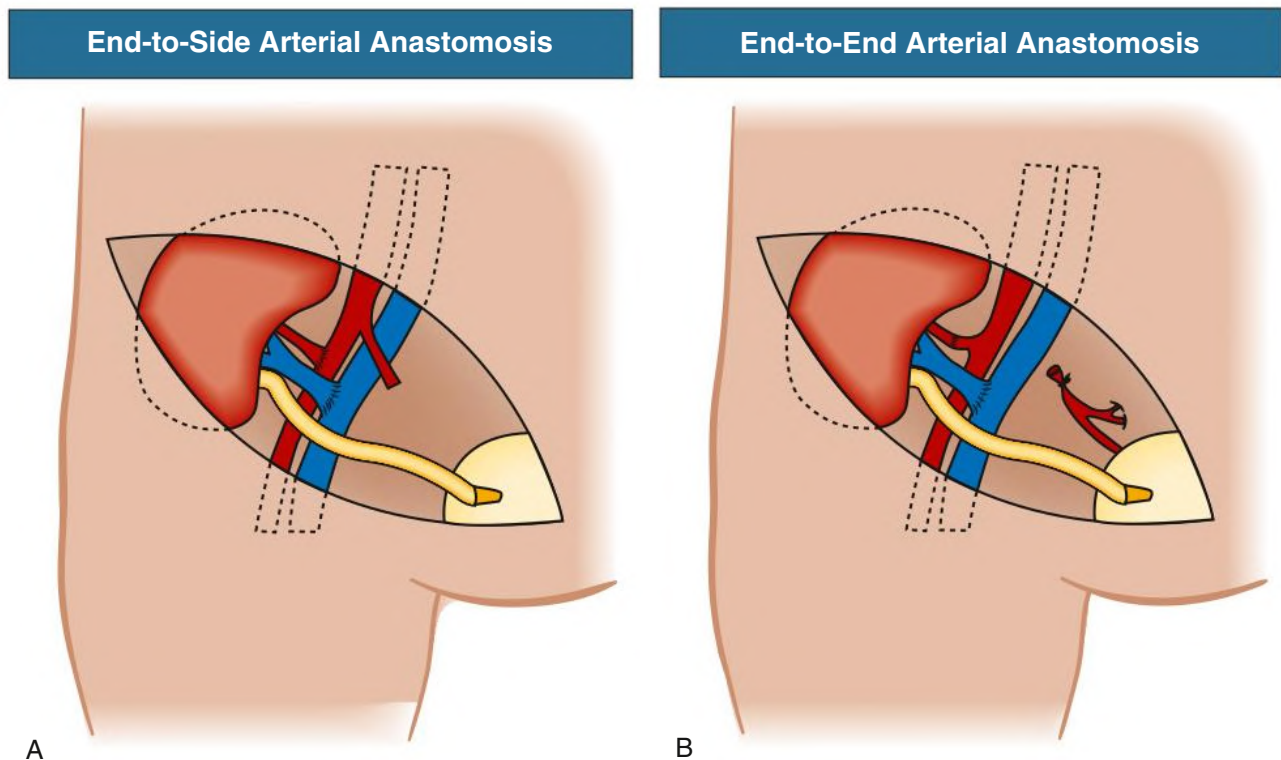
The renal vein is anastomosed end to side to the external iliac vein. The arterial anastomosis can be performed either end to side to the external iliac artery or end to end to the divided internal iliac artery (Fig. 108.5). The end-to-side anastomosis is technically easier and is the usual method used in cadaveric transplantation, where it is possible to include a Carrel aortic patch with the renal artery.

With living donor kidneys it is not possible to include a Carrel patch, and occasionally a cadaveric kidney may be provided without

a useable patch. In these circumstances the options are to anastomose the renal artery end to end to the divided internal iliac artery or end to side to the external iliac artery. Use of an aortic punch to create a circular arteriotomy may facilitate the latter technique. For living donor kidneys, most surgeons use the external iliac artery. It is our experience that positioning the kidney is often easier if the internal iliac artery has been used for the anastomosis, but this does risk buttock claudication and potentially increases the risk for erectile dysfunction in male recipients.

After completion of the vascular anastomoses, the kidney must sit in such a position that the renal vessels are not kinked. The transplanted kidney can be placed laterally in the iliac fossa or may be placed in a subrectus pouch fashioned specifically for the purpose.<sup>22</sup> In the latter case the renal vessels run laterally from the kidney, and this should be noted when a posttransplantation biopsy is performed. An operative diagram of the position of the kidney and vessels is therefore an important component of the clinical notes.

If there are multiple renal vessels, the number of anastomoses should be minimized. This usually can be achieved by careful bench surgery before implantation. If there are two or more renal arteries, their aortic patches are joined in such a way that a single arterial anastomosis is required. If necessary, recipient iliac artery or saphenous vein is used to facilitate reconstruction. Occasionally, small polar arteries will be recognized only after a kidney has been retrieved, and it is particularly important to reanastomose lower polar arteries accurately because these may provide all the ureteral blood supply. In the case of double renal veins, the most common course of action is simply to ligate the smaller vein; the larger one is usually sufficient to drain the whole kidney. If there are two equally sized veins, both may need to be anastomosed, either separately to the external iliac vein or joined and anastomosed together.



**Fig. 108.5** Vascular Anastomosis Techniques for kidney Transplantation. (A) End-to-side anastomosis to the external iliac artery. (B) End-to-end anastomosis to the divided internal iliac artery, suitable for living donor transplantation in which no aortic patch is available.

## Urinary Drainage

The traditional method of ureteral anastomosis is the Politano-Leadbetter technique, involving a transvesical ureteroneocystostomy with creation of a submucosal antireflux tunnel. The end of the transplanted ureter is drawn through a submucosal tunnel from outside to inside and sutured to the bladder mucosa. The majority of surgeons now prefer the technically simpler extravesical ureteroneocystostomy onlay, in which the spatulated end of the ureter is anastomosed to the cystostomy and the divided muscle layer is then resutured over the ureter to create a short antireflux muscle tunnel. The onlay method has the advantage of being possible with only a short length of ureter. The shorter the ureter, the less likely it is that there will be an inadequate blood supply to the distal end, thereby reducing the risks for ischemic ureteral leaks or stenosis. A temporary double-J ureteral stent is usually placed. Stents reduce the impact of small technical errors while the ureter is healing and reduce major urologic complications to an incidence of 1.5%.<sup>23</sup> However, they are a potential source of urinary tract infection, can become encrusted or blocked by debris, and can migrate or fragment. Nevertheless, antibiotic prophylaxis is not justified because it increases the risk for infection with multiresistant organisms. A further danger is the forgotten stent that has not been removed, which should always be considered in patients with unexplained and persistent lower urinary tract symptoms after transplantation. Stents are usually removed 4 to 6 weeks posttransplantation, and this can be performed without general anesthesia with use of a flexible cystoscope.

## Alternative Techniques of Urinary Reconstruction

Kidney transplantation is quite commonly performed in patients who have abnormal bladders. In many patients it is possible to anastomose the transplanted ureter to the bladder in the hope that the bladder can be rehabilitated, if necessary, with use of posttransplantation intermittent self-catheterization. Nonetheless, some patients require urinary diversion with an ileal conduit. The conduit should be fashioned at least 6 weeks before transplantation, but it may have been present for many years. If so, a contrast study (a conduitogram) should be performed before transplantation to exclude the development of conduit stenosis, although this is rare. The transplanted kidney is best placed in the ipsilateral iliac fossa to avoid tension in the ureter, and it may be preferable to deliberately place the transplanted kidney upside down so that the ureter runs cranially and has a more direct route to the conduit. After revascularization, the peritoneum is opened, and the ureter is anastomosed to the conduit over a double-J stent. Excellent long-term results have been achieved with this technique.<sup>24</sup>

## Drainage and Wound Closure

Both the transplant bed and the subcutaneous tissues may be drained to prevent the accumulation of serosanguineous fluid or lymph around the transplanted kidney, although many surgeons no longer routinely place drains. The skin is best closed with a subcuticular absorbable suture and then dressed with a clear adhesive dressing so ultrasound scanning can be performed early without disturbing the wound. For this reason, metal clips are rarely used for the skin.

## Postoperative Course

The recipient is nursed in a general ward with standard precautions and no need for reverse barrier nursing. There is no benefit to continuing prophylactic antibiotics beyond a single dose given at induction of anesthesia. Prophylaxis against thromboembolic disease is in the form of thromboembolic deterrent stockings, and low-molecular-weight heparin (LMWH) is continued until discharge. Oral fluids and diet are commenced as tolerated immediately after surgery. Maintenance

immunosuppressive agents are given orally, except in the presence of a prolonged postoperative ileus with high volumes of nasogastric drainage, when they should be administered intravenously. If recovery is straightforward, the bladder catheter and wound drains are usually removed by day 5, and the recipient is fit for discharge after 7 to 10 days to be kept under close outpatient monitoring.

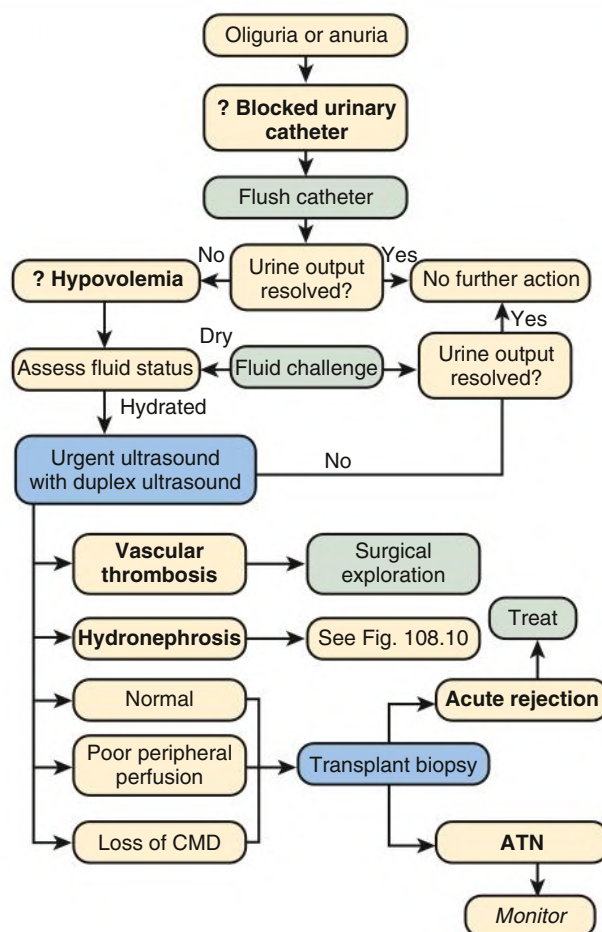
## SURGICAL COMPLICATIONS OF KIDNEY TRANSPLANTATION

There is a small but significant incidence of technical complications, which can be minimized by avoiding damage to the kidneys at the time of the retrieval. Nonetheless, the presence of multiple renal vessels and donor atherosclerotic disease does increase the likelihood of technical problems in the recipient, as do recipient obesity, atherosclerosis, and previous transplantation. Algorithms to aid in the management of complications in the early posttransplant period can be found in Figs. 108.6 and 108.7.

## Wound Infection

The use of preoperative prophylactic antibiotics, commonly amoxicillin-clavulanic acid, has reduced the incidence of wound infection

### Management of Sudden Oliguria or Anuria in the Early Posttransplant Period



**Fig. 108.6** Management of sudden oliguria or anuria in the early posttransplant period. ATN, Acute tubular necrosis; CMD, corticomedullary differentiation.

### Management of Transplant Pain or Swelling in the Early Posttransplant Period

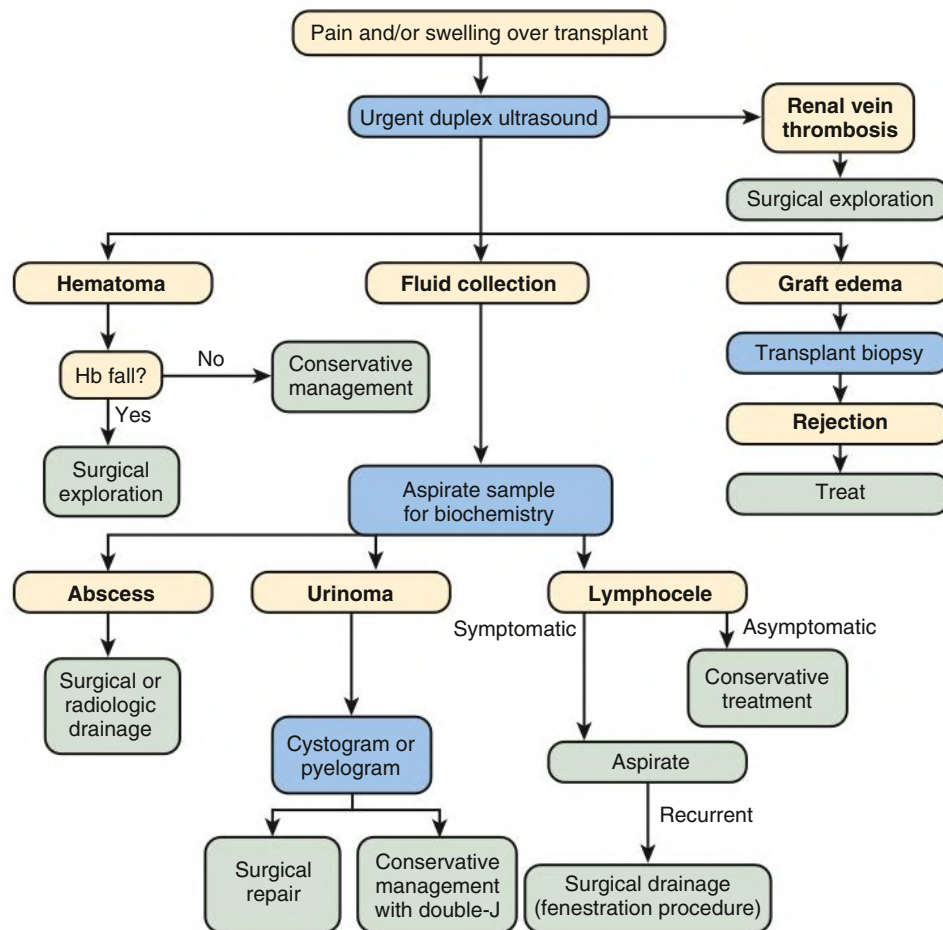


Fig. 108.7 Management of transplant pain or swelling in the early posttransplant period. *Hb*, Hemoglobin.

to less than 1%. If a wound infection does occur, treatment is with antibiotics, guided by microbiologic wound swabs, and drainage of collections as necessary.

#### Wound Dehiscence

The risk for wound dehiscence is increased in obese and diabetic patients and those receiving sirolimus. Identification and treatment of any infection are mandatory. Resuturing of the wound is rarely justified. Large areas of dehiscence often benefit from vacuum-assisted closure, but the majority require only frequent dressing.

#### Vascular Complications

Transplant vascular thrombosis is a feared complication that may cause early and irreversible graft failure. Although there are also significant hemorrhagic risks, routine perioperative prophylaxis should be given with subcutaneous LMWH, and some units prescribe aspirin for the first few postoperative months.

#### Bleeding From Vessels in the Renal Hilum

Careful postoperative observation and regular hemoglobin and hematocrit measurements are crucial for the early detection of bleeding. Output from the transplant drains may give an early indication of heavy blood loss. Unsecured small vessels in the renal hilum may not be obvious during surgery, but they may start bleeding postoperatively.

This form of blood loss can be slow, persistent, and serious. If the patient's condition allows, urgent imaging may be performed to secure a diagnosis, but the best course of action is usually emergency exploration of the transplant under general anesthesia.

#### Anastomotic Hemorrhage

Anastomotic hemorrhage is a rare occurrence, usually caused by a technical surgical error, and is more common with multiple arteries and the use of antiplatelet agents.<sup>25,26</sup> Early after transplantation, the patient may report pain over the graft. This symptom always should be taken seriously. There also may be pain in the back or the rectum caused by a tension hematoma in the retroperitoneum or pelvis. Significant hemorrhage will be attended by circulatory collapse, with tachycardia and hypotension. There will be a decrease in the hemoglobin and hematocrit, sometimes to alarmingly low levels. The patient must be returned to the operating room immediately and the transplant reexplored.

Hemorrhage also can occur some weeks after transplantation because of the development of a mycotic aneurysm of the renal artery. In the rare case of a ruptured mycotic aneurysm, an immediate graft nephrectomy is required, but the mortality is high.

#### Renal Artery Thrombosis

Renal artery thrombosis is a rare event, occurring in less than 1% of transplants. The usual outcome is loss of the kidney. Acute arterial



thrombosis may occur intraoperatively or during the first days or weeks after transplantation. Potential causes include hyperacute rejection or a procoagulant state, but most cases are caused by a technical error during the anastomosis of small or atheromatous vessels.<sup>26</sup> Successful vascular anastomosis requires that the vessels are not under tension and that there is a smooth transition between the two endothelial surfaces; sutures must be placed through all layers of the vessel walls, so an intimal flap is avoided. Vascular adventitia is thrombogenic and must be excluded from the lumen of the anastomosis. The risk for renal artery thrombosis is increased in the presence of atherosclerosis, persistent hypotension, volume depletion (e.g., diarrhea, excessive preoperative dialysis), and prothrombotic states, including diabetes.

Renal arterial thrombosis manifests with sudden anuria, the differential diagnoses being a blocked urinary catheter, dehydration, acute tubular necrosis, or a urologic complication. A high index of suspicion is required to make this diagnosis, particularly in the immediate postoperative period. The only worthwhile investigation is an urgent duplex ultrasound scan, but if the diagnosis is seriously entertained, the only hope of saving the transplant is to reexplore it immediately in the hope that a correctable cause can be found. The reality is that unless the acute arterial thrombosis occurs during surgery, there is little chance of saving the transplanted kidney. Acutely thrombosed grafts must nevertheless be explored and removed to avoid the development of sepsis in a necrotic graft, a potentially fatal complication.

### Renal Vein Thrombosis

Renal vein thrombosis is more common than arterial thrombosis and occurs in 1% to 6% of renal transplants.<sup>26,27</sup> Although it may result from a technical error at the time of surgery, its cause is usually less certain. The renal vein can certainly be twisted or kinked if it is not correctly placed after completion of the vascular and ureteral anastomoses. The peak incidence of renal vein thrombosis is 3 to 9 days after transplantation;<sup>28</sup> transplant patients with good initial graft function will have a sudden loss of urine output, which is often markedly bloodstained, associated with severe pain arising from swelling and (very rarely) rupture of the allograft. The ipsilateral leg may swell if there is involvement of the iliac venous system. Renal vein thrombosis also may be occult and is one differential diagnosis of delayed graft function. Duplex ultrasound scanning is the best investigation. In an established renal vein thrombosis this may show an obviously swollen allograft with surrounding hematoma and an absence of renal perfusion. Lesser degrees of thrombosis, or indeed incipient thrombosis, may be highlighted by an absence of arterial flow in diastole. An even later development is a reversal of flow in diastole.

As with arterial thrombosis, if this diagnosis is entertained, the best course of action is to reexplore the transplant as an emergency. The renal vein anastomosis can be opened to allow the clot to be extracted, and the venotomy is then closed and the kidney observed for improvement. A more radical alternative is to immediately explant the kidney by taking down the arterial, venous, and ureteral anastomoses. The kidney can then be refushed with cold perfusion fluid on the back table and held in preservation fluid at 4°C. This allows much more time to assess the cause of the venous thrombosis, and if the kidney remains viable, the transplant operation can be repeated. If the transplant is already infarcted or cannot be adequately flushed with preservation fluid, the organ will need to be discarded anyway and nothing is lost by immediate explantation. Successful emergency surgical exploration with subsequent long-term function is rare. Interventional radiographic techniques offer an alternative to surgery. The renal vein can be selectively catheterized via the ipsilateral femoral vein, and graft thrombolysis then may be attempted. This technique is particularly useful when renal vein thrombosis occurs late after transplant and the

risk for systemic anticoagulation is low. The use of various thrombolytic agents has been reported, including heparin, urokinase, streptokinase, and tissue plasminogen activator, with no consensus as to which is the most appropriate.

### Transplant Renal Artery Stenosis

Transplant renal artery stenosis is a later complication occurring 3 to 48 months after transplantation. Not all stenoses are of functional or clinical significance, as shown by studies in which all functioning transplants have undergone angiography.<sup>29</sup> Causal factors include donor and recipient atherosclerosis, factors associated with surgical technique, and severe acute rejection.<sup>30</sup> The presentation and management of transplant renal artery stenosis are discussed in [Chapter 43](#).

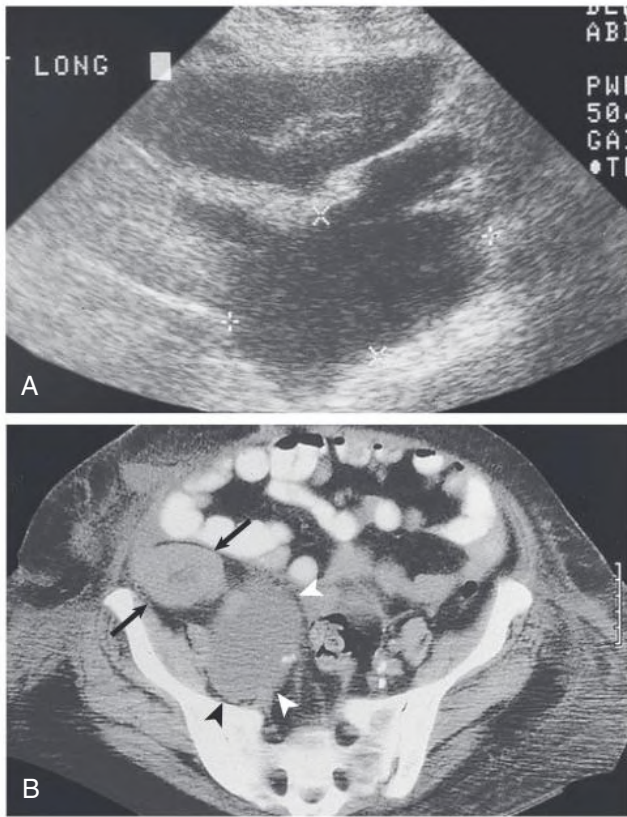
### Lymphocele

Small, clinically insignificant lymphatic collections can be demonstrated by ultrasound scan in up to 50% of kidney transplants.<sup>31</sup> Larger lymphoceles that cause complications or require treatment occur in 2% to 10% of patients.<sup>32</sup> The source of peritransplant lymph leaks is the lymphatic channels around the iliac arterial system rather than the lymphatics of the transplanted kidney itself.<sup>33</sup> Therefore, during the dissection of the iliac arterial system, all the surrounding lymphatic channels must be meticulously secured with nonabsorbable ligatures or metal clips. Wound suction drains should not be removed postoperatively until less than 30 mL of fluid is produced on 2 consecutive days. It is safe to leave drains in place for several weeks posttransplantation to allow a low-volume lymphatic leak to seal by gradual fibrosis. Despite the theoretical risk for infection, this does not seem to be a problem in practice. If necessary, the patient can be discharged from the hospital with the drain in situ.

Compression of the transplanted ureter leading to kidney dysfunction is produced only by very large lymphoceles (volume > 300 mL). The peak incidence is at 6 weeks, but a lymphatic collection may manifest 2 weeks to 6 months after transplantation.<sup>31</sup> Most lymphatic collections are found anterior to the iliac vessels and lying between the transplant and the bladder ([Fig. 108.8](#)). Presenting features may include wound or ipsilateral thigh swelling in association with suprapubic discomfort and urinary frequency caused by bladder compression. Other presentations include pain over the transplanted kidney, sometimes associated with fever, ureteral obstruction with graft dysfunction, and ipsilateral thrombophlebitis. However, the vast majority are asymptomatic and manifest as an incidental finding during an ultrasound scan being performed for another reason. It is important to aspirate all peritransplant fluid collections under ultrasound control to aid diagnosis. Macroscopic findings are usually sufficient to differentiate infected from noninfected lymph, and biochemical analysis of the fluid allows a urine leak to be excluded. Computed tomography or magnetic resonance imaging is an essential investigation if surgery is being contemplated, particularly if a laparoscopic procedure is planned. This allows accurate definition of the relationship between the lymphocele and the transplanted ureter. If the ureter is bow-strung across the superior surface of the lymphocele, it could be damaged during a laparoscopic fenestration procedure.

Many small lymphoceles are asymptomatic and will resolve spontaneously given enough time. If action is deemed necessary, first-line treatment is aspiration under ultrasound control. If there is a recurrence, further aspirations can be performed or an external drain can be placed with ultrasound guidance. If these simple measures fail, open or laparoscopic surgical drainage may be required. A 5-cm-diameter disk of the lymphocele wall is removed to create a large opening into the peritoneal cavity, allowing reabsorption of the lymph through the abdominal lymphatic drainage system. These peritoneal fenestrations





**Fig. 108.8 Posttransplantation Lymphocele.** (A) Ultrasound appearance. A large, echolucent lymphocele can be seen inferior to the transplanted (marked by crosses). (B) Computed tomography appearance. A 5 × 5 cm lymphocele (arrowheads) is present under the transplanted kidney (arrows).

tend to heal before the lymphocele is completely reabsorbed, leading to early recurrence; a metal or omental plug may prevent this.

## Urologic Complications

Urinary tract complications are relatively common after kidney transplantation, with an incidence of 5% to 14%.<sup>34</sup> Although they can be difficult to manage, they only rarely cause graft loss or mortality. The relatively high incidence of urologic problems is a consequence of the tenuous blood supply of the transplanted ureter. After kidney retrieval, the only ureteral blood supply that is preserved is derived from the renal artery near the hilum of the kidney, and this can be easily damaged during retrieval.

### Urinary Leaks

Urinary leaks most commonly occur because of ischemic necrosis in any part of the transplanted urinary collecting system. The distal ureter has the poorest blood supply and is therefore the most common site. Less commonly, leaks occur from the renal pelvis or the midportion of the ureter, which may be a result of unrecognized direct damage to the ureter during organ retrieval. Urinary leaks tend to occur in the first few days after transplantation but can manifest much later. The usual presentation is with straw-colored fluid leaking directly from the transplant wound or accumulating in the drains in association with oliguria. Alternatively, extravasating urine may accumulate as a peritransplant fluid collection. This manifests as a painful swelling of the wound, and the patient may have a fever. In either case, the extravasated fluid must be differentiated from lymph by biochemical analysis of the fluid and a simultaneous serum sample. Urine will have markedly elevated

urea and creatinine levels compared with the patient's serum, whereas lymph will have a similar biochemical profile.

The presence of a urinary fistula should be confirmed by antegrade or retrograde pyelography. Both techniques present challenges. Antegrade puncture of a nondilated pelvicaliceal system is technically difficult but usually possible. Retrograde cannulation of the transplanted ureteral orifice can be attempted with a flexible cystoscope. This is also a difficult maneuver because the transplanted ureter is implanted into the dome of the bladder rather than at its base. If the urine leak is contained as a urinoma, ultrasound will demonstrate a fluid collection between the transplanted kidney and the bladder, which can be sampled by needling or drained by the placement of a suitable percutaneous catheter.

The management of urinary leaks has changed significantly in recent years. The former practice of early reexploration and surgical reconstruction<sup>35</sup> is no longer always necessary. Interventional radiographic techniques offer an alternative, at least for initial treatment. The aim is to place a double-J (pigtail) ureteral stent across the region of damage via an antegrade nephrostomy; this may allow time for the urinary fistula to heal.<sup>36</sup> This technique, however, is unlikely to be successful if there is significant ischemic necrosis of the ureter, in which case surgery still has a role. Reexploration of kidney transplants is straightforward in the early postoperative period but may be a considerable challenge later because of the development of an intense peritransplant fibrotic reaction. The choice of operative procedure for a necrotic distal ureter depends on the length of remaining viable ureter. If there is sufficient length after excision of the necrotic distal portion, the transplanted ureter may simply be reimplanted into the bladder. If this is not possible, the urinary tract should be reconstructed with use of the patient's native ureter. Depending on length of viable transplanted ureter, there is a choice between anastomosing the native ureter to the transplanted ureter proximal to the ischemic segment (uretero-ureterostomy) or to the transplanted renal pelvis (ureteropyelostomy; Fig. 108.9). Whichever technique is chosen, the anastomosis should be protected with a double-J stent. Although these techniques require the native ureter to be ligated proximally, there is usually no need to perform an ipsilateral nephrectomy.<sup>37</sup> Postoperatively, the antegrade nephrostomy can be left in situ so a contrast study can be performed after 7 to 10 days to confirm healing of the new anastomosis. If the transplant recipient has undergone an ipsilateral nephrectomy in the past or the native ureter is too diseased to be used for reconstruction, a Boari bladder flap can be used to reconstruct the urinary tract.

### Ureteral Obstruction

Obstruction of the transplanted ureter may occur at any time after transplantation. It should always be considered in the differential diagnosis of acute transplant dysfunction and excluded by ultrasound examination. The management of transplant ureteral obstruction is summarized in Fig. 108.10. Early obstruction is uncommon and suggestive of a technical error, such as creating a submucosal bladder tunnel that is too tight, kinking of a redundant length of ureter, and incorrect suture placement during anastomosis. Early obstruction may be caused by a blood clot in the ureter, bladder, or catheter. Bleeding may occur from the ureterovesical anastomosis or cystostomy or after a transplant biopsy. It is common practice to drain the urinary bladder using a three-way irrigating catheter because small-diameter two-way Foley catheters are easily blocked by blood clot.

Late ureteral obstruction may occur at the vesicoureteral or pelvi-ureteral junctions. Ischemia that is not severe enough to cause necrosis is presumed to be the cause of most vesicoureteral obstructions.<sup>38</sup> Kidney transplants invariably excite a pronounced perigraft fibrotic response, and this is more likely to be the cause of an obstruction at the

### Ureteral Reconstruction Using Native Ureter

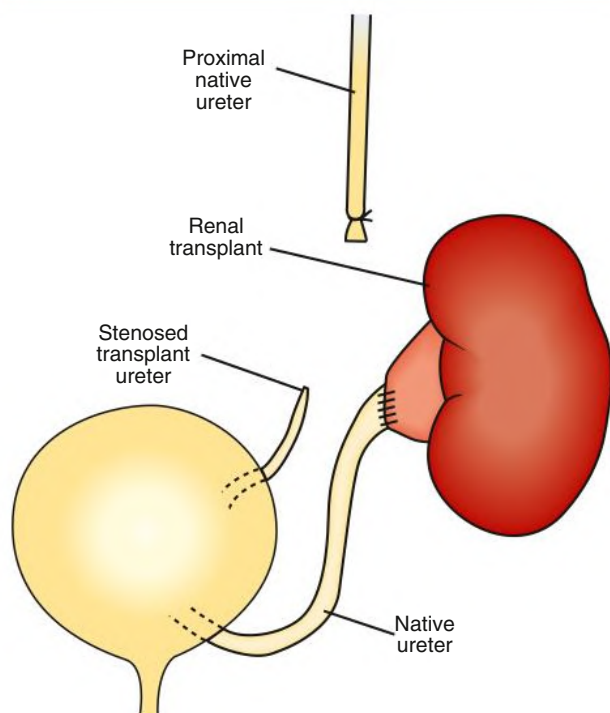


Fig. 108.9 Ureteral reconstruction using native ureter.

level of the pelviureteral junction. It is also possible that acute rejection episodes contribute to subsequent fibrosis.<sup>36</sup> BK polyoma virus also can cause late ureteric obstruction because of hypertrophy of ureteric epithelial cells in combination with infiltration of inflammatory cells.

An ultrasound scan will demonstrate a dilated pelvicaliceal system. However, long-standing kidney transplants may have quite marked pelvicaliceal dilation without being obstructed. This most commonly causes uncertainty in assessment of whether obstruction may be contributing to chronic allograft dysfunction in a patient with biopsy-proven chronic allograft nephropathy. Further investigation is needed to confirm or refute the presence of obstruction and define its anatomy. Retrograde pyelography has a low success rate because of the difficulty of catheterizing the transplanted ureteral orifice at cystoscopy. Therefore, percutaneous nephrostomy followed by antegrade pyelography is the investigation of choice in suspected transplant ureteral obstruction. The nephrostomy is performed under antibiotic cover using ultrasound control, and the nephrostomy tube should be left in place for a few days. If serum creatinine decreases during this period, obstruction is confirmed. If there is no improvement in kidney function, significant obstruction can be confidently excluded. This simple observation avoids the need for an antegrade pressure study (Whitaker test), which may be difficult to interpret in transplanted kidneys. After external decompression of the transplanted kidney for a few days, an antegrade pyelogram is obtained to accurately define the anatomy of the obstructing lesion.

Nonoperative approaches for the treatment of transplant ureteral stricture are often preferred.<sup>39</sup> The simplest approach is to place a double-J stent across the stricture via a percutaneous nephrostomy. This may require initial balloon dilation.<sup>40</sup> The stent can be removed after 6 weeks, but the restenosis rate is high. An alternative is long-term

### Management of Transplant Ureteral Obstruction

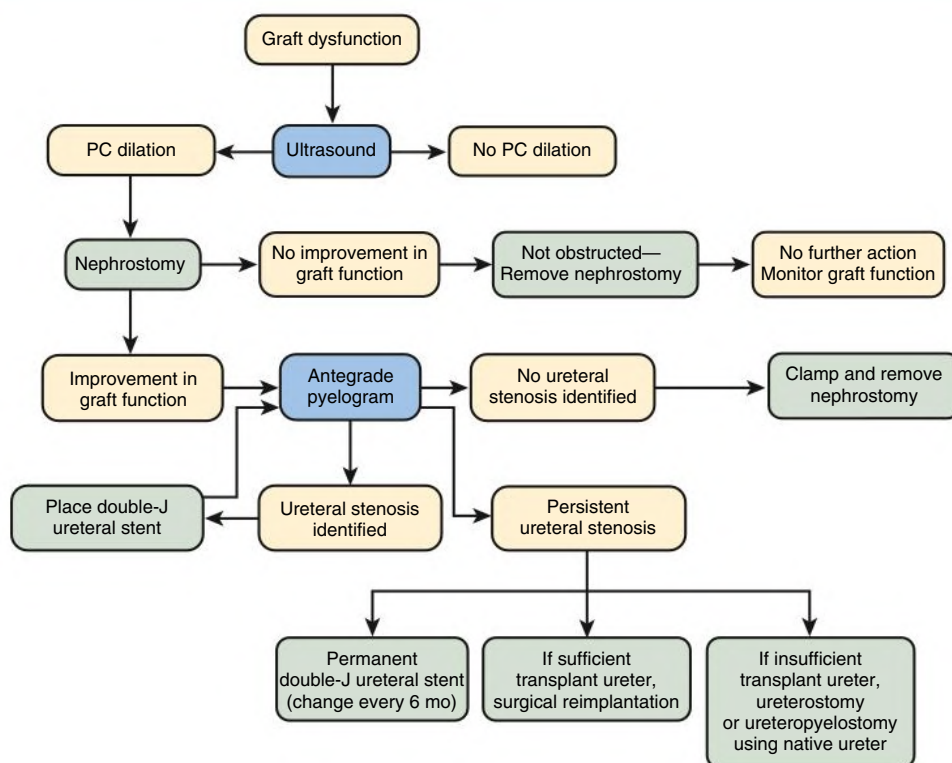


Fig. 108.10 Management of transplant ureteral obstruction. PC, Pelvicaliceal.

stenting, changing the stent every 6 months. The disadvantage of this method is a high incidence of urinary tract infection, with potential severe consequences for immunosuppressed patients, and long-term antibiotic prophylaxis is a sensible precaution. Open surgical management still has a place in the management of ureteral obstruction. The operation performed depends on the site of obstruction and remaining length of healthy transplanted ureter proximal to the obstruction (see discussion of urinary leaks). Not all cases of obstruction require intervention. When there is a mild degree of obstruction not associated with urinary tract infection and in a long-standing kidney that is affected by chronic allograft nephropathy, it may be better to simply monitor transplant function, reserving intervention for a later date should it become necessary.

### Complications in the Transplant Bed

A number of nerves may be encountered in the retroperitoneal dissection required for kidney transplantation. These include the lateral femoral cutaneous nerve and the femoral, obturator, and sacral nerves. Each of these may be damaged by a traction injury, particularly when modern fixed wound retraction systems are used, because these can exert a great deal of pressure on the surrounding tissues. Patients with such neurapraxias should recover completely, but this may take some months, and the condition can be very disabling.

In male transplant recipients, the spermatic cord must be mobilized during the dissection to gain access to the retroperitoneal space. Damage to the testicular artery in the cord can result in testicular atrophy.

### TRANSPLANT NEPHRECTOMY

The optimal management strategy for patients with a failed kidney transplant remains unclear, with a lack of good-quality evidence.

Transplant nephrectomy is mandatory for early graft failure caused by vascular thrombosis, capsular rupture, and irreversible rejection. However, the management of a kidney transplant that has chronically failed is more challenging. The options are transplant nephrectomy or leaving the graft in situ, with or without continuation of immunosuppression. Mortality from both infection and cardiovascular disease has been shown to be higher in patients with failed grafts continuing immunosuppression.<sup>41</sup> However, weaning and discontinuation of immunosuppression also have been shown to increase the risk for transplant nephrectomy and allosensitization.<sup>42</sup>

If the graft remains in situ without immunosuppression, signs and symptoms such as pain, fever, hematuria, and thrombocytopenia may prompt transplant nephrectomy, although the patient also can be treated initially with corticosteroids.

Historically, transplant nephrectomy was advocated to remove antigenic stimulation for anti-human leukocyte antigen (anti-HLA) antibody production, which might adversely affect the possibility of retransplantation. However, there is some evidence that transplant nephrectomy, particularly late after transplantation, may actually increase allosensitization.<sup>43</sup> The suggestion is that the graft may act as an “immunologic sponge” to absorb antibody or may regulate the production of antidonor antibody by the recipient’s immune system.

Early graft nephrectomy is straightforward, but after the first few weeks kidney transplants usually develop quite intense perigraft fibrosis, and this can make late allograft nephrectomy a difficult technical challenge. A subcapsular dissection is preferred, and after removal of the kidney, the hilum is sutured, leaving a cuff of donor vessels in place. Careful hemostasis is required, and the whole raw capsular bed should be cauterized. The wound is usually closed without drains.

## SELF-ASSESSMENT QUESTIONS

- Which of the following statements are *false* regarding organ donation and retrieval?
  - For donation after brain death, the aorta is cannulated after cardiac arrest.
  - Kidneys to be obtained by donation after cardiac death sustain a period of warm ischemia during retrieval.
  - Approximately 30% of kidney transplants in the United Kingdom are currently from living donors.
  - Laparoscopic donor nephrectomy is contraindicated with multiple renal arteries.
  - The usual hospital stay after laparoscopic donor nephrectomy is 1 to 2 days.
- Which of the following statements are *true* regarding the kidney transplant surgical procedure?
  - The kidney is placed in an extraperitoneal position in one or the other iliac fossa.
  - For cadaveric kidney transplants, the renal artery is usually anastomosed to the internal iliac artery.
  - Lower-pole renal arteries can be sacrificed without increasing the risk for ureteral complications.
  - The transplant ureter is anastomosed directly to the bladder.
  - Transplant ureteral stents should remain in place for 3 months.
- Which of the following are *not* risk factors for transplant renal artery thrombosis?
  - Atherosclerosis
  - Dehydration
  - von Willebrand disease
  - Diabetes
  - Protein C deficiency
- Which of the following statements are *true* regarding urologic complications after kidney transplantation?
  - Urologic complications are less common than vascular complications.
  - Urologic complications invariably manifest early.
  - Urine leaks are diagnosed by markedly raised urea and creatinine levels in the fluid compared with serum.
  - Reexploration and surgical reconstruction are always required for urine leaks.
  - Transplant ureteral stenosis is associated with BK polyoma virus infection.
- Which of the following statements are *true* regarding transplant nephrectomy?
  - Transplant nephrectomy is mandatory for early graft failure caused by thrombosis.
  - Transplant nephrectomy reduces allosensitization.
  - Mortality is higher in patients with failed grafts weaned from immunosuppression.
  - Graft nephrectomy, particularly late after transplantation, is straightforward.
  - All failed kidney transplants should be removed.



## REFERENCES

- Wu DA, Watson CJ, Bradley JA, et al. Global trends and challenges in deceased donor kidney allocation. *Kidney Int.* 2017;9:1287–1299.
- Criteria for the diagnosis of brain stem death. Review by a working group convened by the Royal College of Physicians and endorsed by the Conference of Medical Royal Colleges and their Faculties in the United Kingdom. *J R Coll Physicians Lond.* 1995;29:381–382.
- Kootstra G, Daemen JH, Oomen AP. Categories of non-heart-beating donors. *Transplant Proc.* 1995;27:2893–2894.
- Daemen JW, Kootstra G, Wijnen RM, et al. Nonheart-beating donors: the Maastricht experience. *Clin Transpl.* 1994;303–316.
- Hart A, Smith JM, Skeans MA, et al. OPTN/SRTR 2018 annual data report: kidney. *Am J Transplant.* 2020;20(suppl 1):20–130.
- Statistics and Clinical Studies, NHS Blood and Transplant. Annual Report on Kidney Transplantation: Report for 2020–2021. <https://www.odt.nhs.uk/statistics-and-reports/organ-specific-reports/>.
- Foroutan F, Friesen EL, Clark KE, et al. Risk factors for 1-year graft loss after kidney transplantation: systematic review and meta-analysis. *Clin J Am Soc Nephrol.* 2019;14(11):1642–1650.
- Rankin SC, Jan W, Koffman CG. Noninvasive imaging of living related kidney donors: evaluation with CT angiography and gadolinium-enhanced MR angiography. *AJR Am J Roentgenol.* 2001;177:349–355.
- Namasivayam S, Small WC, Kalra MK, et al. Multidetector-row CT angiography for preoperative evaluation of potential laparoscopic renal donors: how accurate are we? *Clin Imaging.* 2006;30:120–126.
- Ratner LE, Ciseck LJ, Moore RG, et al. Laparoscopic live donor nephrectomy. *Transplantation.* 1995;60:1047–1049.
- Nanidis TG, Antcliffe D, Kokkinos C, et al. Laparoscopic versus open live donor nephrectomy in renal transplantation: a meta-analysis. *Ann Surg.* 2008;247:58–70.
- Pace KT, Dyer SJ, Phan V, et al. Laparoscopic v open donor nephrectomy: a cost-utility analysis of the initial experience at a tertiary-care center. *J Endourol.* 2002;16:495–508.
- Nicholson ML, Elwell R, Kaushik M, et al. Health-related quality of life after living donor nephrectomy: a randomized controlled trial of laparoscopic versus open nephrectomy. *Transplantation.* 2011;91:457–461.
- Xiao Q, Fu B, Song K, et al. Comparison of surgical techniques in living donor nephrectomy: a systematic review and Bayesian network meta-analysis. *Ann Transplant.* 2020;25:e926677.
- Ratner LE, Buell JF, Kuo PC. Laparoscopic donor nephrectomy: pro. *Transplantation.* 2000;70:1544–1546.
- Brook NR, Nicholson ML. An audit over 2 years' practice of open and laparoscopic live-donor nephrectomy at renal transplant centres in the UK and Ireland. *BJU Int.* 2004;93:1027–1031.
- Peng P, Ding Z, He Y, et al. Hypothermic machine perfusion versus static cold storage in deceased donor kidney transplantation: a systematic review and meta-analysis of randomized controlled trials. *Artif Organs.* 2019;43:478–489.
- Tingle SJ, Figueiredo RS, Moir JA, et al. Hypothermic machine perfusion is superior to static cold storage in deceased donor kidney transplantation: a meta-analysis. *Clin Transplant.* 2020;34(4):e13814.
- Jochmans I, Brat A, Davies L, et al. Oxygenated versus standard cold perfusion preservation in kidney transplantation (COMPARE): a randomised, double-blind, paired, phase 3 trial. *Lancet.* 2020;396:1653–1662.
- Nicholson ML, Hosgood SA. Renal transplantation after ex vivo normothermic perfusion: the first clinical study. *Am J Transplant.* 2013;13:1246–1252.
- Hosgood SA, Saeb-Parsy K, Wilson C, et al. Protocol of a randomised controlled, open-label trial of ex vivo normothermic perfusion versus static cold storage in donation after circulatory death renal transplantation. *BMJ Open.* 2017;7:e012237.
- Wheatley TJ, Doughman TM, Veitch PS, Nicholson ML. Subrectus pouch for renal transplantation. *Br J Surg.* 1996;83:419.
- Wilson CH, Bhatti AA, Rix DA, Manas DM. Routine intraoperative ureteric stenting for kidney transplant recipients. *Cochrane Database Syst Rev.* 2005;(4):CD004925.
- Abusin K, Rix D, Mohammed M, et al. Long-term adult renal graft outcome after ureteric drainage into an augmented bladder or ileal conduit. *Transpl Int.* 1998;11(suppl 1):S147–S149.
- Osman Y, Shokeir A, Ali-el-Dein B, et al. Vascular complications after live donor renal transplantation: study of risk factors and effects on graft and patient survival. *J Urol.* 2003;169:859–862.
- Hernández D, Rufino M, Armas S, et al. Retrospective analysis of surgical complications following cadaveric kidney transplantation in the modern transplant era. *Nephrol Dial Transplant.* 2006;21:2908–2915.
- Granata A, Clementi S, Londrino F, et al. Renal transplant vascular complications: the role of doppler ultrasound. *J Ultrasound.* 2014;18:101–107.
- Beyga ZT, Kahan BD. Surgical complications of kidney transplantation. *J Nephrol.* 1998;11:137–145.
- Lacombe M. Arterial stenosis complicating renal allotransplantation in man: a study of 38 cases. *Ann Surg.* 1975;181:283–288.
- Bruno S, Remuzzi G, Ruggenti P. Transplant renal artery stenosis. *J Am Soc Nephrol.* 2004;15:134–141.
- Pollak R, Veremis SA, Maddux MS, Mozes MF. The natural history of and therapy for perirenal fluid collections following renal transplantation. *J Urol.* 1988;140:716–720.
- Zincke H, Woods JE, Leary FJ, et al. Experience with lymphoceles after renal transplantation. *Surgery.* 1975;77:444–450.
- Ward K, Klingensmith 3rd WC, Sterioff S, Wagner HN Jr. The origin of lymphoceles following renal transplantation. *Transplantation.* 1978;25:346–347.
- Mundy AR, Podesta ML, Bewick M, et al. The urological complications of 1000 renal transplants. *Br J Urol.* 1981;53:397–402.
- Palmer JM, Chatterjee SN. Urologic complications in renal transplantation. *Surg Clin North Am.* 1978;58:305–319.
- Nicholson ML, Veitch PS, Donnelly PK, Bell PR. Urological complications of renal transplantation: the impact of double J ureteric stents. *Ann R Coll Surg Engl.* 1991;73:316–321.
- Lord RH, Pepera T, Williams G. Ureteroureterostomy and pyeloureterostomy without native nephrectomy in renal transplantation. *Br J Urol.* 1991;67:349–351.
- Brook NR, Waller JR, Pattenden CJ, Nicholson ML. Ureteric stenosis after renal transplantation: no effect of acute rejection or immunosuppression. *Transplant Proc.* 2002;34:3007–3008.
- Goldstein I, Cho SI, Olsson CA. Nephrostomy drainage for renal transplant complications. *J Urol.* 1981;126:159–163.
- Ooms LSS, Moelker A, Roodnat JJ, et al. Antegrade balloon dilatation as a treatment option for posttransplant ureteral strictures: case series of 50 patients. *Exp Clin Transplant.* 2018;16:150–155.
- Smak Gregor PJ, Zietse R, van Saase JL, et al. Immunosuppression should be stopped in patients with renal allograft failure. *Clin Transplant.* 2001;15:397–401.
- Augustine JJ, Woodside KJ, Padiyar A, et al. Independent of nephrectomy, weaning immunosuppression leads to late sensitization after kidney transplant failure. *Transplantation.* 2012;94:738–743.
- Sener A, Khakhar AK, Ngan CY, et al. Early but not late allograft nephrectomy reduces allosensitization after transplant failure. *Can Urol Assoc J.* 2011;5:E142–E147.